

Harry Nguyen

+1 470-667-9000 | harrynguyenswe@gmail.com | linkedin.com/in/harrynguyen26 | github.com/HarryNguyen2662 | timoto.ai

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Profiled settings navigation as p99 bottleneck at 1,200ms using system-level metrics and profiling tools; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec** — selected Protobuf for schema evolution and zero-copy cross-process message passing
- Architected event-driven **TypeScript/React** observer system across **25K+ lines** of Chromium at **95% test coverage**; changes reviewed, merged to production branch, and adopted by senior Chrome engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed **DynamoDB hot-partition failure** at **9,000+ req/sec** using access pattern analysis; redesigned partition key strategy — **30% read throughput improvement** under sustained high-concurrency traffic
- Designed idempotent **REST APIs** with **Redis** distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on **AWS** with JWT-based auth, auto-scaling supporting **500+ concurrent users**; established **CI/CD pipeline** (GitHub Actions) achieving **90% deployment time reduction**; diagnosed N+1 query bottleneck, redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

TiMoto AI – Distributed Backend Platform | *Python, Django, gRPC, REST, AWS, K8s*

timoto.ai

- **Sole engineer** — designed, built, and operate multi-service backend: **gRPC** inter-service layer, Python/Django **REST backend**, PostgreSQL, and multi-AZ **AWS ECS Fargate** with **Terraform** IaC and health-check-triggered auto-rollback
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause via structured logging to circular resource acquisition; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Implemented **circuit breaker pattern** with metrics-driven alerting — **99.9% uptime**, **44% cost reduction** to \$40–60/month; vLLM inference engine with PagedAttention for zero OOM failures under concurrent load

Pulumi – Open Source Contributor | *Go, TypeScript, IaC*

24.4K★ github.com/pulumi/pulumi

- Contributed **Go** CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers; codebase uses distributed state synchronization with **Raft/Paxos** consensus

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, C++, Java, JavaScript, Rust, SQL, Bash

Frameworks & Databases: Django, FastAPI, React, Node.js, REST APIs, PostgreSQL, MySQL, Redis, DynamoDB

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB, Lambda), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Raft/Paxos, Fault tolerance

Observability & ML: Metrics instrumentation, Distributed tracing, vLLM, PagedAttention, LLM-as-a-judge evaluation, PyTorch